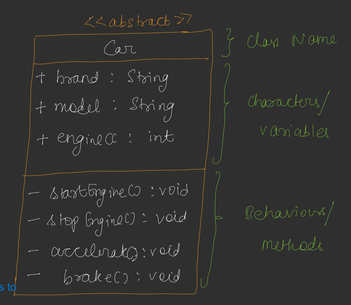
Class Diagram

```
Class Car {
    String brand;
    String model;
    int engineCC;

    startEngine();
    stopEngine();
    accelerate();
    brake();
}
```

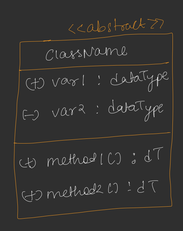


- Unified Modeling Language (UML) is a general-purpose modeling language. The main aim of UML is to define a standard way to visualize the way a system has been designed.

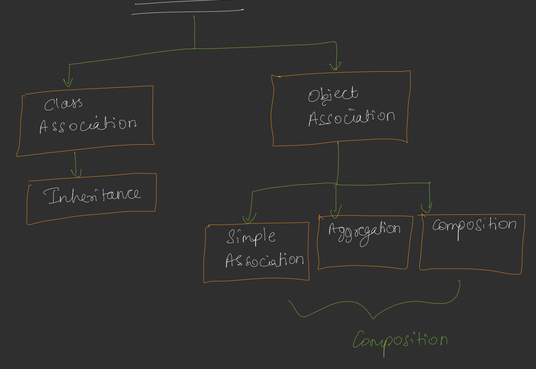
- It is quite similar to blueprints used in other fields of engineering. UML is not a programming language, it is rather a visual language.

Generic Class

Class diagrams to shows the static structure of a system by showing system's classes, their methods and attributes. class diagrams also help us identify relationship between different classes or objects.



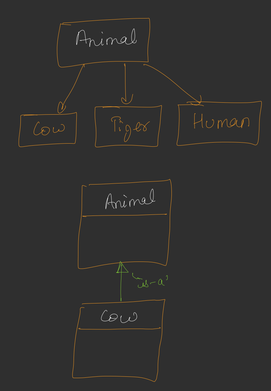
Associations



Inheritance

- (is-a) relationship
- Denoted by (⟶)

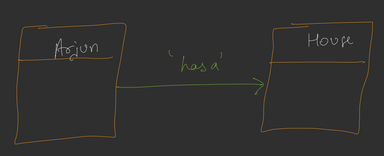
```
Class A {
    method1();
}

Class B : Public A {
    method2();
}

main() {
    B* b = new B();
    b->method1();
    b->method2();
}
```



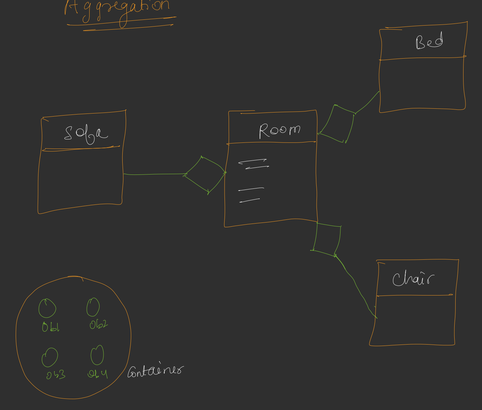
Composition

- (has-a) relationship
- Denoted by (⟶)

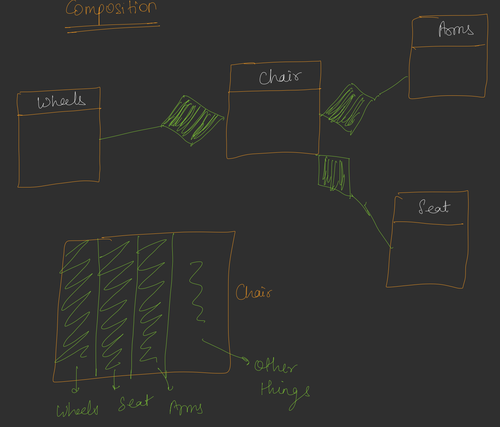
Simple Association



Aggregation



Composition



Composition in Code

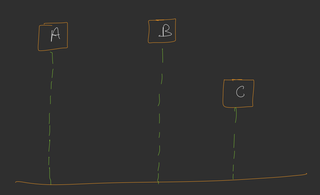
```
Class A {
    method1();
}

Class B {
    A * a;
    B() {
        a = new A();
    }
    method2();
}

main() {
    B * b = new B();
    b -> method2();
    b -> a -> method1();
}
```

A sequence diagram simply depicts interaction between objects in a sequential order i.e. the order in which these interactions take place.

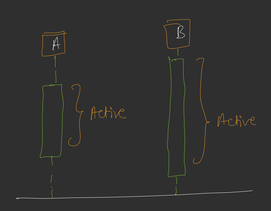
Sequence Diagram

① Representing Object:

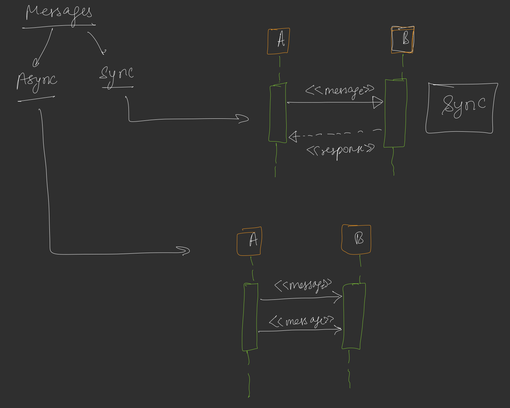
② Lifeline



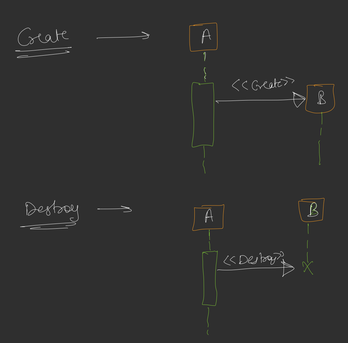
③ Activation Bar



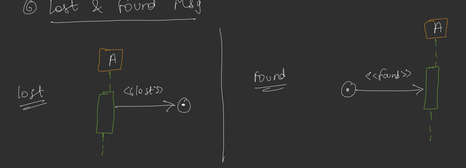
④ Messages



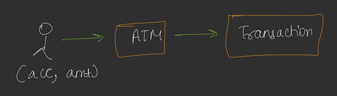
⑤ Create & Destroy Message



⑥ Lost & Found Msg



ATM Transaction flow



Use Case

1.

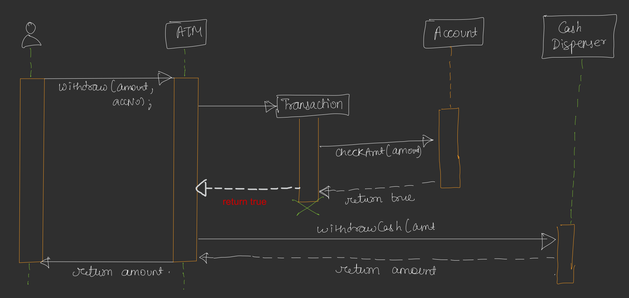
2.

① Verify ATM pin
② verify Account
③ Tell Cash Dispenser for money.

Sequence Diagram

① Use Case
② Objects
- ATM
- User
- Transaction
- Account
- Cash Dispenser

③ Draw Diagram



alt : (if-else)
Option : (if)
loop : for/while